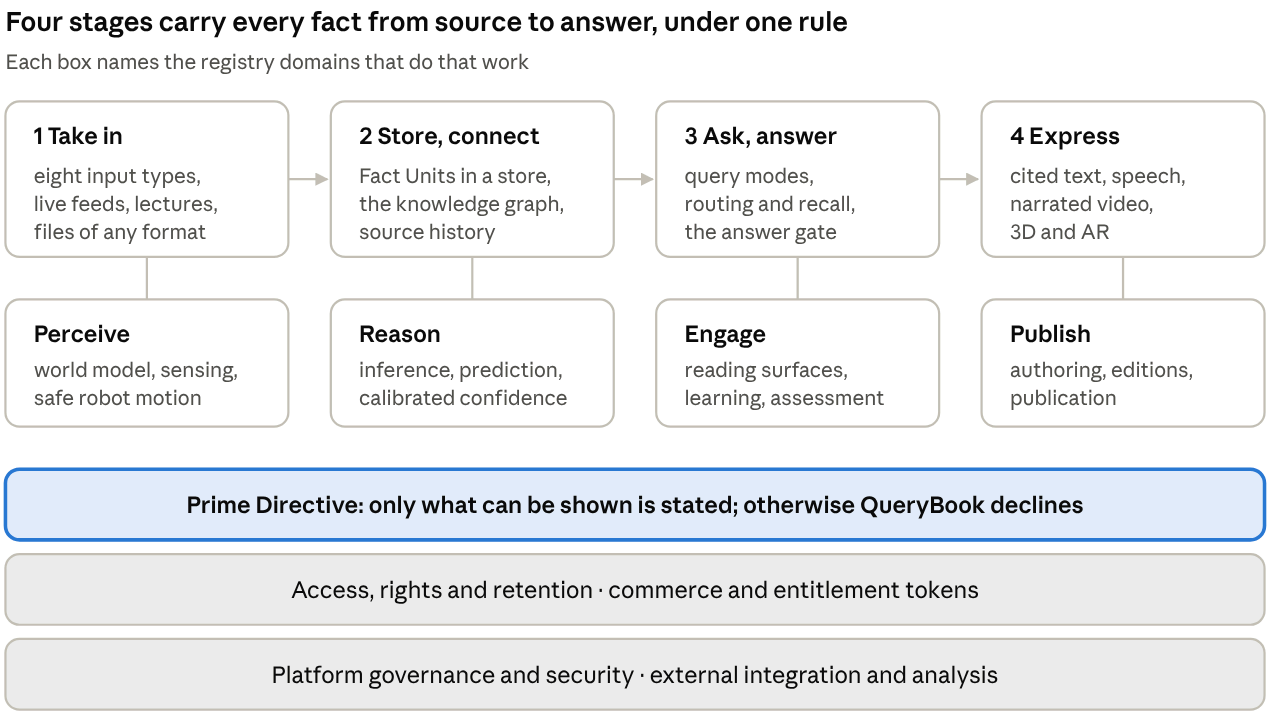
QueryBook Executive Overview: Every Feature, Illustrated

Nov 8, 2026 · @Garry Howard

QueryBook turns any source — text, images, audio, video, data, live feeds — into verified, sourced facts, and answers questions only from those facts. This overview covers all 407 features in the QueryBook Composite Feature Registry (v71), grouped into 15 domains, each described in plain words.

How QueryBook works

Every fact moves through four stages — take in, store and connect, ask and answer, express — and every stage obeys one rule: state only what can be shown.



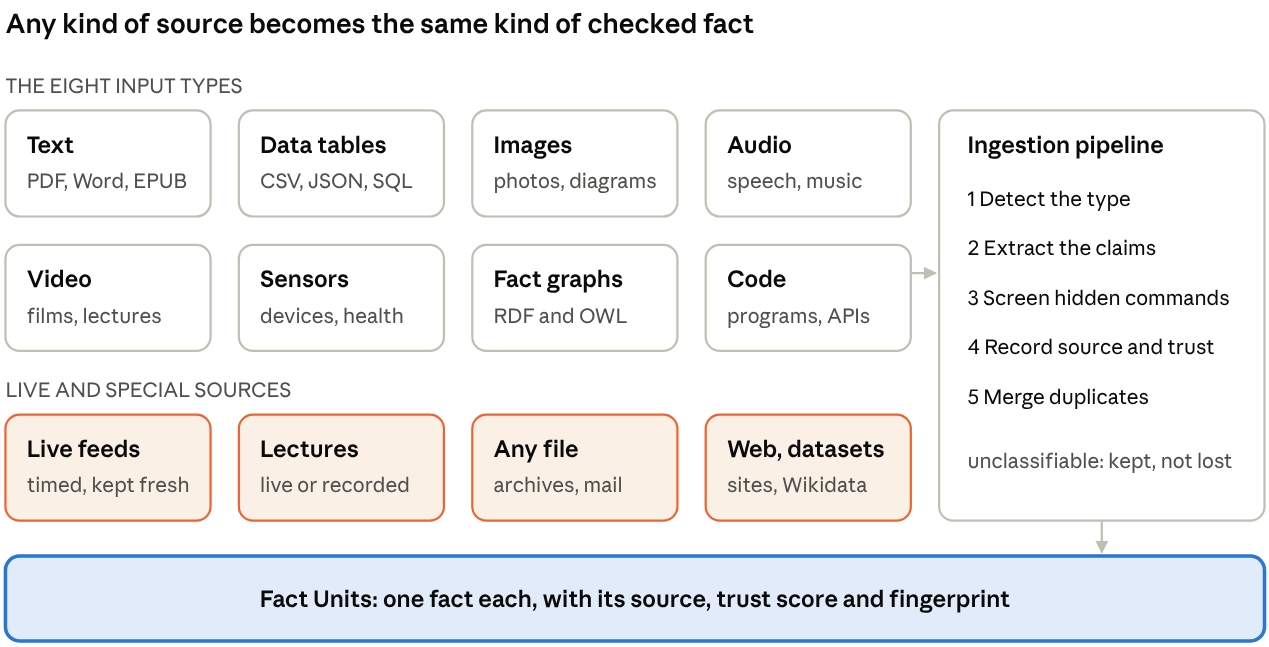
how QueryBook works · the 15 registry domains in one picture

A **Fact Unit** is one fact as a sealed record: the fact itself (subject, relationship, object), its source, a trust score from 0 to 1, and a fingerprint that is the same wherever the same fact appears. Facts live in the **fact store** (UFCS, the Universal Fact Content Substrate), indexed for millisecond lookup and connected by a **knowledge graph** of concepts and a **source history** that records where everything came from. Answers are built only from

facts above the trust threshold, each one cited; when trusted sources disagree, both are shown; when nothing trusted is found, QueryBook says so. Around this core sit the capabilities that perceive and reason, engage readers and learners, publish, protect rights, and govern the platform — the 15 domains below.

What QueryBook takes in

QueryBook ingests eight types of content plus live feeds, lectures and files of any format, and turns all of them into the same thing: checked Fact Units.



ingestion · eight input types plus live and special sources

Input type	Examples
Unstructured text	Web pages, PDF, Word, EPUB, plain text
Structured data	Database tables, CSV, TSV, JSON
Images	Photographs, diagrams
Audio	Speech recordings, music, environmental sound
Video	Raw video, screen recordings, lectures
Sensor data	Device telemetry, biometric streams

Input type	Examples
Knowledge graphs	RDF and OWL, property graphs
Code	Source code in all major languages, API specifications

Live feeds. A live data source (prices, readings, statistics) is registered with its publisher, refresh interval, how old a value may get before it is treated as stale, its unit, and whether it may be redistributed. A feed missing any of these can be connected but is never used to answer questions.

Lectures and talks. A live or recorded talk is captured with consent, transcribed, split by speaker and stored with each claim attributed to the person who said it.

Any file. Documents, spreadsheets, presentations, archives, messages, images, audio and video are accepted. Archives are opened within strict size limits, programs are described but never run, and hidden instructions inside content are screened out.

Languages and voices. Vocabulary, pronunciation and grammar are acquired per language from lawful sources and a governed bilingual dictionary; voiceprints are enrolled only with consent.

How people ask: the interactive query modes

People can ask, explore, compare documents, study, translate or present, and every mode answers only from Fact Units, citing them or declining.

Six families of query modes, all held to one rule: cite or decline

Ask and check

Standard Q&A with citations
Follow-up questions in context
Clarifying questions
Claim checks, contradictions

Explore

Semantic exploration
Timeline reconstruction
Discussion mode
Fact graph views

Many documents

Session books across a library
Edition comparison
Collaborative reading
Lecture records, cited

Learn and retain

Tutorial mode
Comprehension checks
Flashcards, spaced repetition
Progress and certificates

Language and voice

Translation and detection
Pronunciation and speech
Spoken questions
Voice navigation

Present and program

Tours and presenter mode
Immersive 3D, AR overlays
Structured queries
Plain-language programs

Every mode answers from Fact Units only: VERIFIED, CONTRADICTED or UNKNOWN

interactive query modes · six families

Mode	What it does	Registry No.
Standard Q&A	Answers from verified facts found with enough confidence, each one cited	1, 362
Guided exploration	Answers follow-up questions using the conversation so far, recent turns weighted most	2
Clarifying questions	Asks a specific question when a request is ambiguous, instead of guessing	68
Claim verification	Lets outside services check a claim and get back verified, contradicted or unknown, with citations	364, 365
Discussion mode	Discusses a book, taking up a viewpoint only when the source supports it strongly enough	3
Semantic exploration	Explores related ideas outward from a starting topic, showing only strong connections	6

Mode	What it does	Registry No.
Timeline reconstruction	Orders a book's dated facts into a timeline	7
Multi-document sessions	Opens a temporary session across a reader's library without copying it, deleted when the session ends	28
Edition comparison	Shows what was added, removed or changed between two editions	8
Collaborative reading	Several readers share a session, each seeing only what all are entitled to	165
Lecture questions	Questions answered from recorded talks, every claim attributed to its speaker	353, 354
Tutorial mode	Turns a book's facts into lessons ordered from easier to harder	4
Comprehension check	Asks a question from a fact and marks the learner's answer	5
Retention (flashcards)	Builds cited flashcards and schedules them by spaced repetition for long-term memory	9, 113
Progress and certification	Tracks mastery without ranking learners; certificates need a passing score and identity check	169, 172
Translation and language	Translates, detects languages, pronounces and speaks, shows languages side by side and maps idioms, grounded in stored language facts (prototype Language page, suite Language Builder)	64, 145, 184, 345, 355
Voice navigation	Moves through content by voice, with revocable consent and no stored voiceprint	167

Mode	What it does	Registry No.
Tours and presenter mode	Guided tours through scenes, and presentation modes the author has authorised	24, 25, 26
Immersive 3D and AR	3D scenes built only when the facts hold enough spatial detail; cited facts overlaid on a camera view	23, 166
Structured queries and programs	Precise read-only queries, and plain-language instructions broken into checked steps	19, 20

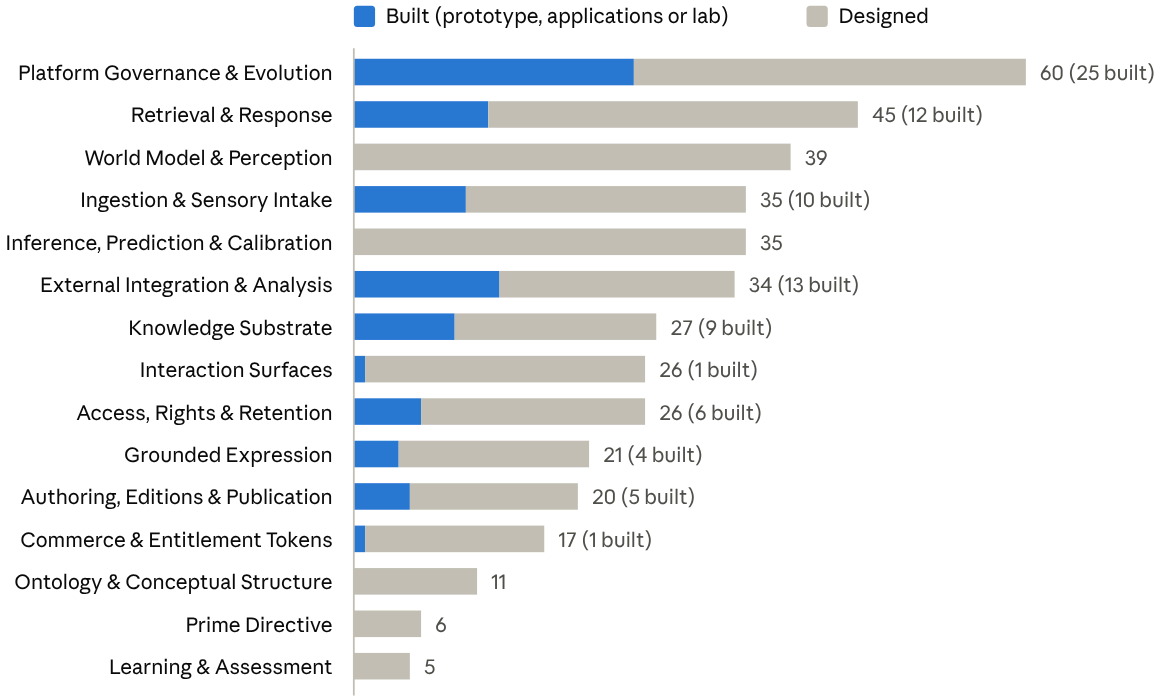
Retention also has a second meaning in QueryBook: session records are kept by default, with erasure available at any time (registry No. 131, under Access, Rights & Retention below).

The registry at a glance

The registry lists 407 features in 15 domains; governance, retrieval, perception and ingestion are the largest.

Features by domain, built and designed

407 features in 15 domains; 86 built so far



QueryBook Composite Feature Registry v71 · 407 features

How to read the status column. *Built* means the registry records the feature as working in the QueryBook 9.64 prototype, the suite applications, a laboratory prototype or a test tool. *Designed* means it is fully specified in the registry and the QueryBook Bible. The registry has recorded build status only for features added since version 66, so some *Designed* features (Standard Q&A, for example) also run in the prototype under a later, *Built* entry.

The tables that follow list every feature, by domain, with its registry number.

Prime Directive

The safety rules every other feature obeys: answers and ingestion run on separate paths, safety comes first in every decision, and nothing is stated that cannot be shown. 6 features, 0 built.

No.	Feature	What it does	Status
56	Safety Substrate	Checks every operation against safety rules at required checkpoints, letting it proceed only when it passes every applicable check.	Designed
176	Domain Transfer Matrix	Controls which parts of the system may pass information to which others, blocking by default, so commerce and analysis never influence answers.	Designed
177	Query/Ingestion Path Separation	Keeps the fast question-answering path largely separate from background data intake, so neither slows the other down.	Designed
178	Domain Slice Priority Ordering	Schedules work by system area, putting safety first and commerce last, so commercial content is dropped rather than delaying an answer.	Designed
195	Interpretive Composition Order	Applies five presentation adjustments in a fixed, predictable order, with safety review before and after, and user-stated preferences overriding inferred ones.	Designed
255	Context Lock and Key	Attaches a key describing the context of each output, so identical keys guarantee identical results and differences signal a defect.	Designed

Ingestion & Sensory Intake

How content of every kind enters QueryBook — the eight input types, sensory devices, live feeds, lectures, outside AI replies and files of any format — and is screened before any fact is kept. 35 features, 10 built.

No.	Feature	What it does	Status
51	Multimodal Ingestion Pipeline	Identifies which of eight defined content types incoming material belongs to, choosing the type with the strongest match.	Designed
52	Classification Fallback	Matches unclear incoming content to the nearest content type when confidence is sufficient, and otherwise stops with a stated reason instead of guessing.	Designed
74	Sensory Abstraction Pipeline (SAP)	Converts raw sensory input into structured, concept-level facts before storage, so no raw recordings are admitted to the knowledge store.	Designed
106	Sensory-Annotated Content Playback	Plays sound, visual or device vibration effects only for content an author has tagged with them, with sound and vibration requiring user opt-in.	Designed
118	External Sensory Hardware (Smell/Haptic)	Triggers external scent or touch hardware only when a user deliberately acts on author-defined content, never from signals the system infers, like emotion.	Designed
125	Ego-Motion Compensation for Non-Stationary Capture	Corrects for camera movement when analyzing video, subtracting the camera's own motion so objects and scenes are detected more accurately.	Designed
127	Intersensory Redundancy Weighting	Gives more confidence to a detected property when several input types confirm it at the same moment than when only one does.	Designed

No.	Feature	What it does	Status
128	Object Permanence — Precise Correspondence	Keeps track of an object as the same item when it is briefly hidden during a session, without tracking it across separate sessions.	Designed
146	Linguistic Resource Registry	Records the license terms of every outside language resource before use, consulting it only when its license covers the intended purpose.	Designed
147	Derived-Artifact-Only Extraction	Keeps only statistical summaries, such as word frequencies, from certain licensed language resources, storing or reproducing none of their actual text.	Designed
257	Substrate Output as a Modality	Registers specialized computing hardware, such as neuromorphic processors, as a separate input type that can fail without disrupting other inputs.	Designed
258	Registered Substrate Interpretation	Treats output from specialized computing hardware as evidence about that hardware only, unless a pre-registered interpretation links it to another subject.	Designed
259	Substrate Stimulation Dispatch	Sends stimuli to registered computing hardware only through declared, logged actions, never on the system's own inference, leaving living-tissue oversight to institutions.	Designed

No.	Feature	What it does	Status
264	Engine-Specific Prompt Composition	Builds prompts for each AI model consistently from knowledge records, so identical requests produce identical prompts and differences in replies reflect the model.	Designed
265	Reply Conversion to Knowledge Records	Breaks AI model replies into individual statements, keeping only those that pass format checks, each tagged with its source prompt and model.	Designed
266	Derivation Marking and Disclosure	Permanently marks statements generated by an AI model and discloses this in every citation, so readers know a model, not an author, produced them.	Designed
267	Self-Corroboration Bar	Counts repeated answers from the same AI model as one source, so asking a model repeatedly cannot inflate confidence in a claim.	Designed
268	Citation Verification of a Reply	Checks every citation in AI-generated statements against stored records, marking each verified or unverified, and flags conflicts with authored sources for reviewers.	Designed
270	Specialized Feed Registration	Registers live data feeds with seven required details, such as refresh rate and units, before they can be used to answer questions.	Designed

No.	Feature	What it does	Status
271	Publication Authority Admission	Records published values, such as prices, exactly as the publisher stated them without adding invented uncertainty, while forecasts carry their publisher's stated uncertainty.	Designed
272	Staleness Bound	Serves live data only while it is fresh enough, otherwise reporting its age and publication time instead of giving an outdated answer.	Designed
273	Restatement Distinct From Correction	Separates a publisher's restatement of past figures from an error correction, keeping restated figures valid and referring out-of-policy revisions to a person.	Designed
274	Redistribution Entitlement	Controls what users may do with live data values, such as display or redistribute them, removing restricted values from responses and stating the restriction.	Designed
294	Open Claw Arbitrary-Artifact Ingestion	Accepts files of any format, from documents to video, and converts them into candidate facts, treating their contents strictly as data, never instructions.	Built
295	Content-Based Format Detection	Identifies a file's true format from its contents rather than its name or extension, so renamed files are classified correctly.	Built
296	Archive Expansion Bounds	Checks compressed archives against limits on size, nesting and file count before opening them, refusing oversized archives entirely rather than partially.	Built

No.	Feature	What it does	Status
297	Artifact Segmenter with Source Locators	Splits files into addressable pieces, such as pages, spreadsheet cells or time ranges, so every derived fact cites its exact source location.	Built
298	Executable Described, Never Executed	Records embedded macros, scripts and other executable parts of a file by type and size only, never running or following them.	Built
299	Instruction-Like Content Screen	Screens incoming content, including hidden text, for passages aimed at AI systems rather than people, flagging them so they are never obeyed.	Built
300	Untrusted-Record Trust Floor	Stores flagged suspicious content with a trust score below the answer threshold, so it remains auditable but is never used for answers.	Built
302	Page-Description Text via Character-to-Unicode Maps	Recovers readable text from PDF and similar files using their built-in font mapping tables, so characters appear correctly rather than as codes.	Built
347	LIL Voiceprint Enrollment and Matching	Enrolls and matches voiceprints only with recorded, revocable consent, confirming a person's identity only after a trust check is passed.	Built
353	Lecture Query Ingestion	Captures talks with consent, transcribes them by speaker, and stores each claim as something the speaker said, never as established fact.	Designed

No.	Feature	What it does	Status
355	Deterministic Multilingual Acquisition	Learns each language's vocabulary, sounds and grammar from lawfully sourced texts and a vetted dictionary, withholding any pronunciation that cannot be verified.	Built
367	External Triple Extraction Interface	Lets authorized outside callers turn a sentence into a structured fact, returning sentences it cannot convert as unconverted rather than guessing.	Designed

Knowledge Substrate

The foundations: the Fact Unit format, the store, the source history, offline synchronisation and the integrity checks that keep every fact traceable and tamper-evident. 27 features, 9 built.

No.	Feature	What it does	Status
50	Fact Unit Schema	Gives every stored fact the same fixed set of fields, whatever its source, with an identifier derived consistently from content, source and time.	Designed
53	Provenance Ledger	Logs every consequential action in a tamper-evident record, so the system can detect when a required safety check was skipped or altered.	Designed
61	QueryBook Format (QBF)	Stores facts in a file format where damaged data costs only one small block, so the rest of the file stays readable.	Designed

No.	Feature	What it does	Status
62	Storage Architecture	Places frequently and recently used facts in faster storage and indexes facts for both exact lookups and searches for similar items.	Designed
80	Offline Synchronization Layer	Lets the system work offline and, on reconnecting, reconciles any differences between the offline and online copies of the knowledge.	Designed
281	Epistemic–Aleatory Uncertainty Decomposition	Separates uncertainty that more evidence could reduce from natural variability that cannot be reduced, so evidence gathering stops when it can no longer help.	Designed
282	Assertion Polarity Field	Marks every fact as affirmative or negative, so a confirmed 'no' counts as an answer while missing information still leads the system to decline.	Designed
283	Native N-ary Assertion Structure	Stores a complex statement, with details such as value, unit, method and observer, as a single record rather than several linked fragments.	Designed
284	Applicability Scope Field	Records the conditions under which a fact holds, such as time period or jurisdiction, so facts under different conditions are not treated as conflicts.	Designed
285	Baconian Adjustment History Layer	Keeps a permanent history of every confidence adjustment, so the current confidence in any fact can be traced back to its first observation.	Designed

No.	Feature	What it does	Status
286	Cryptographic Integrity Chain	Links each version of a record to the previous one using standard fingerprints, making any tampering detectable without blockchain or tokens.	Designed
287	Canonical Compact Encoding	Saves each fact in a compact, lossless form that is the same record as the full version, supporting small devices and offline use.	Designed
288	Record-Fused Canonical Encoding	Uses the compact format as each fact's own form in memory, on disk and in transit, so records are readable without an external schema.	Designed
289	Opaque-Blob Storage Contract (Database-Neutral Boundary)	Lets any database store facts as sealed data it simply saves and returns unchanged, so no particular database product or format is required.	Designed
292	Fact Unit Language	Expresses every fact in a formal, language-neutral form, so the same statement in different human languages is recognized as one statement.	Designed
343	Post-Quantum Composite Ledger Signatures	Signs the provenance record with both a classical and a quantum-resistant method, so its trust chain stays verifiable against future quantum computers.	Designed
361	Version-Independent Provenance-Continuous Store	Keeps the knowledge store in a fixed location across software upgrades, so an update never wipes it or replaces it with an empty one.	Built

No.	Feature	What it does	Status
368	Assertion-Type Label (Fact / Opinion / Plan / Prediction)	Labels each candidate record as a fact, opinion, plan or prediction, allowing only records labelled as facts to be certified or cited as verification.	Designed
374	Fingerprint Set Reconciliation (Have / Want)	Lets connected systems compare lists of what facts each holds before syncing, sending only missing facts and rechecking each one on arrival.	Designed
399	Digital Synapse Record	Keeps a detailed record of each working relationship between two agents, including its strength, outcomes and recent changes, so each relationship can be explained.	Built
400	Synaptic Plasticity	Strengthens links between agents when they succeed together and weakens them when they fail, so one coincidence never establishes a relationship.	Built
401	Forgetting, Turnover and Consolidation	Gradually fades and eventually removes unused links between agents, more slowly for proven ones, and reports all such changes.	Built
402	Co-Activation Episodes, Windows and Directed Links	Links agents that work together in one episode, succeed close together in time, or use each other's output, forming links from successes only.	Built
403	Engrams per Context	Remembers which agents were involved in solving each kind of problem, recording only successes and forgetting the least recently used contexts first.	Built

No.	Feature	What it does	Status
404	Functional Agent Assemblies	Groups agents with strong working links into teams, recording members, cohesion, joint successes and the kinds of problems each team solves.	Built
405	Networks of Networks (Assembly Links)	Connects teams of agents through strong links between their members, reporting the number, average strength and contexts of those connections.	Built
406	Spreading Activation over Synapses	Identifies which other agents are likely relevant by following strong links from active agents, used for ordering fallbacks but never changing answers.	Built

Ontology & Conceptual Structure

The knowledge graph of concepts and relationships: how categories inherit properties, how exceptions override rules, and how causes are checked against time. 11 features, 0 built.

No.	Feature	What it does	Status
54	Ontology Mesh	Links facts in the knowledge graph only where a real, named relationship exists, not merely because terms often appear together.	Designed
65	Cross-Domain Rule Engine (CDRE)	Applies rules across subject areas and keeps a running list of contradictions whenever two active rules reach incompatible conclusions.	Designed

No.	Feature	What it does	Status
73	Conceptual Memory Layer (CML)	Keeps stable, general concepts in long-term memory, separate from short-term working context, keeping the concept but not the specific episode behind it.	Designed
112	Cognitive Map Formalization	Applies an existing, predictable indexing method to finding related concepts in the knowledge graph, connecting it to a known neuroscience model.	Designed
141	Concept Attribute Inheritance	Lets specific concepts inherit properties from broader categories when looked up, so corrections to the broader category automatically flow through, with appropriately reduced confidence.	Designed
142	Exception Override Resolution	Lets a specific concept's own stated property override an inherited one, preferring the closest source and reporting unresolved ties as contradictions.	Designed
143	Inverse Edge Traversal	Allows relationships to be read in reverse only where a reverse meaning has been formally declared, instead of guessing at it.	Designed
153	Event Type Catalogue	Keeps a registered catalogue of event types with required fields, admitting an event record only when every required field is present.	Designed
154	Environment Vector Normalization	Stores each snapshot of real-world conditions once and shares it across facts, so corrections reach every fact that refers to it.	Designed
155	Causal Edge Temporal Validation	Accepts a claim that one thing caused another only when the cause did not begin after the effect, logging rejected claims as contradictions.	Designed

No.	Feature	What it does	Status
226	Difference-Preserving Alignment	Merges editions or timelines only where differences are minor, preserving real or unclear differences so readers never lose a genuine fact.	Designed

Retrieval & Response

How questions are understood and answered: the interactive query modes, structured queries and programs, the networks that settle on the best facts, and answer routing. 45 features, 12 built.

No.	Feature	What it does	Status
1	Standard Q&A	Answers a question by finding relevant verified facts, settling on the best set, and composing a reply only from facts retrieved with enough confidence.	Designed
2	Guided Exploration	Answers follow-up questions using the conversation so far, giving recent turns more weight than older ones so replies stay on track.	Designed
3	Discussion Mode	Lets readers discuss a book, taking up a viewpoint only when the source material supports that viewpoint strongly enough.	Designed
4	Tutorial Mode	Turns a book's facts into step-by-step lessons ordered from easier to harder, so readers learn in a sensible sequence.	Designed

No.	Feature	What it does	Status
6	Semantic Exploration	Lets readers explore related ideas outward from a starting topic in the knowledge graph, showing only sufficiently strong connections.	Designed
7	Timeline Reconstruction	Builds a timeline from the dates recorded in a book's facts, putting events in chronological order for the reader.	Designed
8	Edition Comparison	Compares two editions of a book and shows what was added, removed or changed between them.	Designed
9	Flashcard Generation	Creates study flashcards, each drawn from a single verified fact and citing its source, so every card is traceable.	Designed
19	QBQL Structured Query	Lets users run precise structured queries against the facts that only read data and never change anything.	Designed
20	NLPL Natural-Language Programs	Breaks a plain-language instruction into its separate parts before any program is built, stopping rather than guessing when something is ambiguous.	Designed
55	Neuromorphic Cognitive Layer / Attractor Network	Runs the network that settles on the best set of facts, repeatedly adjusting candidates against their neighbours until the result stabilizes.	Designed
57	Semantic Transport Layer	Routes facts between internal components so each component receives only facts matching every filter it has declared.	Designed

No.	Feature	What it does	Status
66	QBQL (execution floor)	Runs structured queries as read-only lookups that cannot change data or carry hidden state between runs.	Designed
67	QBPL (bounded program layer)	Limits user programs to a small fixed set of operations with hard loop limits, so programs cannot run unbounded.	Designed
68	Clarification Generation Engine	Asks the user a specific clarifying question when a request is genuinely ambiguous or the user seems frustrated, instead of guessing.	Designed
69	Natural Language Interaction Layer (NLIL)	Adjusts wording, format and depth of explanation to suit the audience, without ever changing the underlying fact.	Designed
70	Fact Universe Architecture	Defines how the fact graph, knowledge graph and source history fit together as coordinated views of one shared fact store.	Designed
71	Adaptive Response Engine (ARE, pipeline)	Analyzes the facts retrieved for a question and fixes how the response will be shaped before any wording is generated.	Designed
139	Attractor Load Regime Governor	Prevents the network that settles on the best set of facts from being overloaded, refusing new patterns beyond a safe capacity limit.	Designed
223	Energy Descent and Termination	Guarantees the network that settles on the best set of facts always finishes and reaches the same result from the same starting point.	Designed

No.	Feature	What it does	Status
231	Convergence Regime Disclosure	Records which stability guarantee each network operates under and discloses it with every output; only an operator can change it.	Designed
250	Edge-Class Traversal Cost	Limits how far a search spreads by assigning fixed costs to different kinds of connections, recorded with each response and never self-adjusted.	Designed
251	Propagation Reach by Class	Sets how far a search can reach through each kind of connection from its cost, so the same query always gathers the same material.	Designed
256	Relevance Gradient	Steers searches toward areas where relevance changes gradually, prioritizing stable meaning without changing what can be reached or storing extra data.	Designed
260	Analog Convergence Realization	Lets the network that settles on the best set of facts run on analog circuits, where one physical settling replaces many digital calculations.	Designed
261	Unique-Equilibrium Precondition	Allows analog hardware only when the network is configured to have a single guaranteed outcome, so manufacturing differences cannot change results.	Designed
262	Cross-Array Agreement	Requires two analog circuits given the same input to agree within tolerance, raising a fault on disagreement instead of averaging results.	Designed

No.	Feature	What it does	Status
263	Cross-Array Agreement Under Variation	Uses a configuration that keeps differently manufactured circuits agreeing far more often, a finding recorded from simulation rather than hardware tests.	Designed
269	Competitive Candidate Resolution	Weighs competing interpretations of a query against each other at once, and asks for clarification when no single interpretation clearly wins.	Designed
290	Internal Fact Query with Edge Translation	Keeps its internal fact queries separate from outside data stores, translating requests at the boundary so its guarantees never depend on them.	Designed
330	Framed UFCS-FQL Message Format	Packages records and media into checked, optionally signed message frames, dropping any frame that fails integrity or signature checks.	Built
331	Per-Type Compression	Compresses batches of records efficiently and passes media in its standard format, never enlarging or altering compressed files.	Built
332	Control, Normal and Bulk Priority Lanes	Sends control, normal and bulk traffic on separate channels so large transfers never delay urgent messages or live streams.	Built
333	Resynchronizing Frame Reader	Checks each incoming data frame and skips damaged ones while continuing to read the rest of the stream.	Built

No.	Feature	What it does	Status
334	Streaming Package Rebuild with Manifest Check	Rebuilds received media into standard streaming formats and checks its file list, refusing entries that point outside the intended location.	Built
352	PIL Human-Query Interpretation and Ambiguity Gate	Works out what a user's question means and asks one clarifying question when intent remains materially uncertain, rather than guessing.	Built
354	Forever-Linked Lecture Record	Links each lecture record to its speaker, venue, event and organization, citing and attributing claims, or declining when the speaker is unknown.	Designed
362	Cited Q&A / ChatBot Surface	Answers typed or spoken questions only from verified facts, citing each one or honestly saying unknown; no language model asserts facts.	Built
364	Claim Verification Endpoint	Lets authorized outside services check claims, returning verified, contradicted or unknown with citations, without changing stored facts or treating unknown as false.	Designed
365	Contradiction Query Template	Provides a ready-made search for facts that contradict a claim, counting a rival only when it is more trusted than the claim's source.	Designed
386	Reasoning-Aspect Self-Description	Describes the system's reasoning capabilities at several levels of detail, with each status taken from current evidence rather than fixed text.	Built

No.	Feature	What it does	Status
389	Rule-Based Self-Question Answering with Sources	Answers questions about the system itself using fixed rules and cited evidence, giving the same answer every time without a language model.	Built
392	Three-Way Answer Routing (Central, Shadow, DSN)	Lets administrators choose how bulk fact lookups search the stores: Central asks every store (the default), the Digital Synaptic Network follows learned routes, and Shadow compares the two.	Built
393	Learned Store Pathways per Lookup Kind	Learns which data stores tend to answer each kind of lookup, asking productive ones first while storing routes only, never facts.	Built
394	Complete-Answer Fallback	Asks every remaining store when the usual ones return nothing, so smarter routing never hides a fact a full search would find.	Built

Interaction Surfaces

Everything a reader sees and touches: the reader, previews and footnotes, notes and annotations, boards, 3D scenes and tours, AR, voice, and presentation adapted to device, culture and setting. 26 features, 1 built.

No.	Feature	What it does	Status
10	Fact Graph Visualization	Draws the facts and their connections as a visual graph, showing only what the requester is permitted to see.	Designed

No.	Feature	What it does	Status
11	Inline Fact Previews	Shows a quick summary of a cited fact when the reader hovers or taps, without running a new search or revealing uncited material.	Designed
12	Contextual Footnotes	Adds extra detail about a fact as a footnote at the point it is cited, which readers can dismiss without changing the answer.	Designed
13	Personal Notes	Lets readers keep private notes that only they can see, stored alongside the book's facts.	Designed
14	Social Annotations	Lets readers share their annotations with others according to sharing settings the author of each note chooses.	Designed
15	Message Boards	Places each message board post in one of four board types based on the poster's role, and checks moderation rules when posting.	Designed
23	Immersive 3D Scenes (IQR)	Builds a viewable 3D scene only when the underlying facts contain enough spatial detail to support it.	Designed
24	Tour and Exhibition Mode	Lets authors arrange 3D scenes into a guided tour with a fixed order that readers step through at their own pace.	Designed
25	Presenter Mode	Checks that an author has given current, unrevoked permission for a specific presentation mode before that mode is used.	Designed

No.	Feature	What it does	Status
26	Instructor-Created Tours	Lets instructors assemble tours from published scenes, with a permanent label showing the tour was made by an instructor, not the author.	Designed
27	Instructor Notes	Lets instructors write notes that only students on their class roster can see.	Designed
28	Session Books	Creates temporary reading sessions that point to the reader's library without copying it, and deletes the session when it ends.	Designed
38	Class Boards	Provides message boards limited to a specific instructor and class, so posts are visible only within that group.	Designed
103	Native QueryBook Reader	Displays standard electronic books correctly as a normal reader would, adding QueryBook features without breaking standard reading behaviour.	Designed
165	Collaborative Reading	Lets several readers share a reading session, showing only material every participant is entitled to see, so no one gets an unopenable citation.	Designed
166	AR Overlay Presentation	Overlays cited facts on objects in a live camera view, never tags people or animals, and collects no attention or gaze data.	Designed
167	Voice Navigation	Lets users navigate by voice with revocable consent, discards the audio after transcription, keeps no voiceprint, and asks when unsure.	Designed

No.	Feature	What it does	Status
168	Spoiler Protection	Hides facts beyond the reader's current place in a story, saying an answer lies ahead without revealing it.	Designed
182	Media and Immersion Interfaces	Adds media and immersive interfaces that transform content or describe scenes with provenance, withholding output rather than inventing detail when information is insufficient.	Designed
190	Context Modulation	Adapts results to the reader's device, network and setting without changing which records they are entitled to, disclosing any edition fallback.	Designed
191	Perspective Modulation	Frames the same permitted records differently depending on the role a user chooses, such as student or author, without granting extra access.	Designed
192	Ergonomic Modulation	Adjusts layout, density and readability to the reader's pace and accessibility needs, always honouring stated preferences over inferred ones.	Designed
193	Cultural Modulation	Adapts examples and conventions to a locale the user or operator declares, never guessing culture from names, addresses or behaviour.	Designed
194	Social Modulation	Adjusts pacing and shared annotation for classes and group reading, showing only what every participant may see and granting no extra access.	Designed

No.	Feature	What it does	Status
293	Contextual Value Models	Stores declared cultural, social and comfort preferences as versioned settings that shape presentation only, never which facts are used.	Designed
363	Fact-Unit Compressed Search Engine and Deployment Surfaces	Offers the same verified facts through e-books, online services, search and offline analysis, each bound to cite sources or answer unknown.	Built

Learning & Assessment

Study, practice and proof of learning, built only from verified content. 5 features, 0 built.

No.	Feature	What it does	Status
5	Comprehension Check	Generates a question from a verified fact and marks a learner's answer correct when its meaning is close enough to that fact.	Designed
29	Instructor-Customized Assessment Templates	Lets instructors save a reusable quiz structure, while question content still comes from verified facts and question count stays within a set maximum.	Designed
113	Study and Learning Tools	Builds glossaries, study guides, tutorials and spaced-repetition flashcards from already-verified content, grouping related concepts using the knowledge graph.	Designed
169	Learner Progress Tracking	Tracks learner progress and mastery against verified facts, limits instructor access to time-bound class grants, and never ranks or predicts learner ability.	Designed

No.	Feature	What it does	Status
172	Certification	Issues certificates only when learners both meet the score threshold and verify their identity, allowing outside verification without revealing their answers.	Designed

Grounded Expression

How answers are put into words, speech, other languages and media without adding anything the facts do not support. 21 features, 4 built.

No.	Feature	What it does	Status
60	Narrative Graph	Links facts together to shape how answers are composed, and flags gaps where no supporting fact connects the story.	Designed
64	Multilingual Regeneration Engine (MRE)	Regenerates answers in another language from the underlying facts rather than translating text, checking that the meaning survives a round trip.	Designed
79	Structural Groundedness Enforcement	Blocks any statement in an answer that cannot be traced back to at least one supporting verified fact.	Designed
107	Audio Narration	Uses the author's own recorded narration when available, otherwise a generic synthetic voice, and never imitates a real person's voice.	Designed
114	Content Verification and Argumentation Tools	Checks claims against verified facts, shows existing supporting counts, and builds arguments only from sourced records, citing any contradictions found.	Designed

No.	Feature	What it does	Status
119	Source Framing Signals	Shows a source's documented editorial position and measurable citation facts with references, without issuing its own judgment of bias.	Designed
144	Wording-Operative Span Preservation	Reproduces quotations and wording-sensitive passages exactly as written, adds a marked translation aid, and withholds the original where rights do not permit.	Designed
145	Idiom Detection and Equivalence Mapping	Spots idioms when content is added and translates their meaning using registered equivalents, or renders them literally with a note.	Designed
184	Parallel Multilingual Presentation	Shows a work in two or more languages side by side, aligned fact by fact, with gaps shown rather than filled.	Designed
200	Expressive Register Selection	Chooses one approved tone of voice per answer, such as formal or plain, without changing facts, confidence levels or required caveats.	Designed
201	Affective Register Gate	Uses a supportive tone only when the user asks, a setting requires it, or consented voice signals indicate it, never by guessing emotions.	Designed
202	Media-Integrated Expression	Includes images and other media within answers only when a cited verified fact references them, delivered with that fact's citation.	Designed

No.	Feature	What it does	Status
203	Sentence Aggregation	Combines related statements about the same subject into one sentence only when they are similarly well supported, keeping each citation.	Designed
204	Referring Expression Selection	Uses full names at first mention and shorter references afterward, reverting to full names whenever a shorter reference could be ambiguous.	Designed
205	Connective Emission	Adds linking words such as 'because' or 'however' only when the facts record that relationship, avoiding unsupported connections between statements.	Designed
206	Given-Before-New Ordering	Orders sentence parts so already-mentioned information comes before new information, where the target language allows and quotations stay untouched.	Designed
207	Fingerprint-Bounded Lexical Variation	Varies wording with registered synonyms only when the meaning provably stays the same, never altering quotations, official names or specialist terms.	Designed
345	Advanced Multilingual Pronunciation Ladder	Finds pronunciations by trying a built-in tool, then language rules, and returns nothing rather than inventing one when neither applies.	Built
346	Pronunciation and Speech Quality-Control Battery	Tests pronunciation and speech output for each supported language and reports any language that fails rather than quietly serving it.	Built

No.	Feature	What it does	Status
351	PIL Paralinguistic Interpretation	Measures tone and emotional intensity in a voice recording and labels them as estimates, never presenting them as facts.	Built
356	Dialect Parameter Clusters	Identifies a speaker's dialect by matching declared speech and spelling features, using it to shape expression only, never which facts apply.	Built

Authoring, Editions & Publication

Tools for authors and publishers: manuscript upload and editing, coherence checks, collaboration, editions, accessibility, print and analytics. 20 features, 5 built.

No.	Feature	What it does	Status
21	Application Library and Generator	Matches a plain-language request to an existing application template when the match is close enough, and builds a new one otherwise.	Designed
22	Video Colorization & Upscaling (VCUM)	Colorizes and sharpens video by inferring plausible colors from materials and lighting, staying consistent with the source and giving repeatable results.	Designed
31	Manuscript Upload	Takes in an uploaded manuscript in one step and keeps it marked as a draft, hidden until the author formally submits it.	Designed
32	Manuscript Editor	Lets authors edit manuscripts with every change saved as a new numbered version, keeping a full edit history.	Designed

No.	Feature	What it does	Status
33	Narrative Coherence Checking	Checks a draft for internal contradictions, flagging places where two facts state incompatible things about the same subject.	Designed
34	Counterfactual Exploration (Drafting)	Lets authors explore a 'what if' alternative from their current draft while keeping both the original and alternative versions.	Designed
36	Multi-Author Collaboration	Detects when two co-authors have edited the same material at the same time, so conflicting changes are caught.	Designed
37	Author-Hosted Reader Events	Lets authors schedule reader events through a registered outside platform, without giving that platform wider authority over the system.	Designed
39	Publisher Submission and Print-on-Demand	Accepts titles for publishing and print-on-demand, but only after the rights have been validated.	Designed
40	Market Analytics	Combines a title's distribution and usage data into standard reports, with totals summed across regions.	Designed
43	Accessibility Suite	Produces accessible versions of content at the same time as the original, and treats content as unfinished until that version exists.	Designed
75	VCUM Semantic Colorization Engine (SCE)	Infers likely colors for images lacking reliable color, using materials, lighting and historical references, with the same result every time.	Designed

No.	Feature	What it does	Status
76	VCUM Agent Specifications	Defines the specific task of each automated agent in the video restoration process, so no agent acts outside its assigned job.	Designed
134	Author Portal Book Upload	Receives books from verified authors through the web portal, checks rights, and starts processing only after rights are cleared.	Designed
171	Multi-Edition Print Layout	Lays out each print edition from the same underlying facts after checking print rights, keeping page numbers stable and accessible versions included.	Designed
335	Record-Grounded Colorization Confidence Bound	Assigns colors only from recorded material information, with confidence limited by the weaker of image recognition and the record's trustworthiness.	Built
336	Colorization Withhold Threshold	Leaves areas uncolored when confidence is too low, preferring a visible gap to an unsupported color.	Built
337	Luminance-Preserving Colorization with Consistency Check	Adds color without altering original brightness, marks output as reconstructed, checks fidelity, and leaves skin uncolored unless a record supplies the color.	Built
357	Grounded Scene Compilation (Scene Director)	Turns verified facts into checked instructions for an outside video generator, marking returned footage as reconstruction, never as a source.	Built

No.	Feature	What it does	Status
359	Scene Reconstruction from Prose (Scene Reconstructor)	Turns written passages into a sequence of video shots, showing unidentified people as silhouettes and marking the result as reconstruction.	Built

Inference, Prediction & Calibration

Reasoning beyond stored facts, held to strict limits: hypotheses kept apart from facts, forecasts with calibrated confidence, and questions prepared in advance. 35 features, 0 built.

No.	Feature	What it does	Status
46	Human Reasoning and Motivation Modeling	Records people's goals and motivations tied to a named person, and trusts motivations it infers less than ones the person states directly.	Designed
58	Predictive Inference Engine (PIE)	Projects how knowledge is likely to change by combining several input streams, giving a range of likely outcomes rather than a single value.	Designed
59	Scenario Graph	Explores what-if scenarios by recording only the changes from current knowledge, and drops branches whose probability falls below a set threshold.	Designed
63	AGI Extension Modules (AEM)	Combines several stored facts into a new insight when no single fact answers a question, and marks that insight as less certain.	Designed

No.	Feature	What it does	Status
111	Hypothesis Generation	Generates new hypotheses marked as proposed, which only an identified human expert can mark as validated; no automatic approval path exists.	Designed
115	Manifold-Based Trend Forecasting	Forecasts trends from time-series data with an explicit confidence range, shown in a clearly labeled area separate from verified answers.	Designed
123	Boltzmann-Style Diverse Hypothesis Sampling	Optionally generates several varied candidate explanations for an input, labels them as randomly generated, and keeps them as proposals awaiting human validation.	Designed
135	Probable Question Generation (PQG)	Anticipates likely questions when facts are added, before anyone asks, and keeps only questions that the stored facts answer well enough.	Designed
136	Pre-Computed Answer Set	Prepares answers to anticipated questions in advance, traceable to their source facts and no more confident than the least certain fact used.	Designed
137	Question Index Reuse and Invalidation	Serves a prepared answer only when the question matches closely and its source facts are unchanged; otherwise it answers from live retrieval.	Designed
140	Sampling Criticality Control	Lets an operator set how varied generated hypotheses are, records who set it, and keeps those hypotheses out of answers entirely.	Designed

No.	Feature	What it does	Status
150	Attestation Horizon Constraint	Ensures that when testing past forecasts, only sources documented as created before the event are used, excluding any source with an uncertain date.	Designed
151	Retrospective Forecast Scoring	Tests the system's forecasting by predicting past events from earlier information only, then scores accuracy and calibration against what actually happened.	Designed
152	Calibration Subject Class Gate	Restricts forecast-accuracy records to institutions, processes, systems or groups, and refuses to create any such record about an individual person.	Designed
156	Bayesian ConfidenceScore Posterior Update	Updates confidence in a fact as supporting or contradicting evidence arrives, so well-established facts shift less with each new piece of evidence.	Designed
157	Prior-Strength Resistance Check	Makes well-established concepts resist change from a single contradiction, requiring contradicting evidence from multiple sources before they are revised.	Designed
158	Precision-Weighted Error Propagation	Scales how much a contradiction revises a concept by how reliable the contradicting evidence is compared with how settled the concept already is.	Designed
159	Low-Precision Error Suppression	Ignores contradictions too weak to justify any change, but logs them so a pattern of small errors against one item remains visible.	Designed

No.	Feature	What it does	Status
179	Undiscriminated Hypothesis Detection	Flags when two well-supported explanations for the same subject are too close to tell apart, and sends weakly supported cases for more evidence.	Designed
180	Discriminating Instance Selection	Picks the observation that would best tell two competing explanations apart, and states in advance what each possible result would mean.	Designed
187	Reasoning Mode Classification	Stores only conclusions that follow logically from verified records; speculative conclusions are offered as proposals only and never added to the knowledge store.	Designed
188	Inductive Support Threshold	Proposes a general rule only when enough supporting examples exist and supporting evidence outweighs contradicting evidence, keeping weak contradictions on record.	Designed
189	Observation Diversity Measure	Measures how varied the sources behind a fact are, gives less weight to repeated confirmation from one source type, and discloses heavy concentration.	Designed
196	Response Feedback Admission	Records every answer delivered and lets readers report problems, which lower confidence in cited facts gradually rather than overriding or deleting them.	Designed

No.	Feature	What it does	Status
197	Outcome Pairing and Subject Gate	Pairs answers with what actually happened afterward, but refuses to build such records about individual people, only about institutions and groups.	Designed
198	Delivered-Answer Calibration Score	Measures how well stated confidence matched real outcomes, separately by subject area and confidence level, so weak areas are not hidden by averages.	Designed
199	Confidence Band Calibration Gap	Flags confidence levels where stated confidence and real outcomes diverge, sending suggested fixes to a named human reviewer rather than applying them automatically.	Designed
246	Predictive Level Registration	Organizes the system's expectations into four ordered levels, from words within a record up to general rules about the world.	Designed
247	Lowest-Level Error Attribution	Assigns a prediction error to the lowest level that explains it, so a misread sentence never causes changes to the system's understanding of physics.	Designed
248	Escalation Gate	Revises higher-level understanding only when every simpler explanation at lower levels has been ruled out, so core rules change only as a last resort.	Designed

No.	Feature	What it does	Status
249	Unattributed Error Retention	Logs errors no level can explain without changing anything, surfaces repeated patterns for review, and never lets observations override physical laws.	Designed
252	Tri-Temporal Posture	Labels every answer as describing the documented past, the live present or a projected future, and never uses projections as verified evidence.	Designed
253	Six-Dimensional Coherence	Rates how consistent information is across six aspects, such as meaning, timing and evidence, scoring by the weakest aspect so problems are not hidden.	Designed
254	Coherence Classification	Classifies information as stable, drifting or misaligned, naming the weakest aspect and noting missing data or conflicts, without affecting answers or confidence.	Designed
378	Principal Evidence Pair (Agent Reputation Store)	Tracks a reliability score for each automated agent from signed verdicts; a low score can only propose restrictions, which require an operator's decision.	Designed

World Model & Perception

Understanding the physical world: geometry and colour, objects and places, and the safety rules for machines that move near people. 39 features, 0 built.

No.	Feature	What it does	Status
44	Precision Geometric/Color Registration	Compares colors in a presenter's reference material against a set tolerance and reports any drift for review instead of correcting it automatically.	Designed
45	Geometric Proof Engine	Derives new geometric relationships from stored facts using fixed rules, cites each proof step, and stops when it finds a contradiction.	Designed
72	World Aware System Model (WASM)	Keeps a model of the physical world, covering objects, their relationships, physics and possible uses, stored and searchable like any other fact.	Designed
88	Robot Memory Domain (RMD)	Turns robot and sensor data into structured knowledge and sends it back to the robot, without keeping the raw sensor recordings.	Designed
97	World-View Model (WVM)	Maintains the system's own underlying structure for categories, causes and timelines, so its reasoning needs no outside model to express them.	Designed
105	Personal Object Location	Helps people find their belongings on a consented live camera feed, recognizing only objects and materials and discarding any people or animals.	Designed
116	Geometric Shape Matching / Grid-Cell Indexing	Matches shapes by comparing outlines against stored geometry, using a fixed grid so that searching for nearby candidates gives repeatable results.	Designed

No.	Feature	What it does	Status
117	Dual-Grounded Developmental World Model (The Box Principle)	Learns concepts by matching spoken words with seen objects, treating a single pairing as provisional until it has been confirmed through repeated encounters.	Designed
122	Biased Competition for Object Salience	Ranks how prominent each detected object in a scene is, by confidence and task relevance, without ever deleting objects ranked low.	Designed
124	Developmental Parallel — Verified Correspondence	Links the system's word-and-object learning methods to published child development research, stating openly where the match is exact and where it falls short.	Designed
126	Encyclopedia Content Ingestion	Extracts individual facts from encyclopedia text without copying the original wording, and checks accompanying images separately for usage rights.	Designed
129	Additional Disambiguation Heuristics for Cross-Modal Binding	Uses common-sense cues such as gaze direction to help pair words with objects, but always lets direct observation override those cues.	Designed
160	Physical Constraint Enforcement	Rejects any predicted scene that breaks the laws of physics, preferring an answer with an acknowledged gap over one that is physically impossible.	Designed

No.	Feature	What it does	Status
161	Constraint Residual Computation	Checks predicted values against physical laws such as gravity and logs each rejection, so a wrongly configured rule shows up through its rejection rate.	Designed
174	Hyperedge Attention	Weights each fact in a linked group by its reliability and how well other evidence supports it, instead of counting all facts equally.	Designed
175	Attractor-Based Hyperedge Completion	Fills gaps in a partly observed shape by settling on the most consistent complete form, accepting the result only when support is strong enough.	Designed
213	Commandable Equipment Registration	Registers camera, lighting, drone and inspection equipment with its movement and force limits, and refuses to command anything unregistered or any vehicle.	Designed
214	Session Authorization for Actuation	Grants equipment control per session, tied to an approving operator, specific devices, space and expiry, and stops immediately when access expires or is revoked.	Designed
215	Pre-Emission Constraint Veto	Checks every equipment command against safety limits before sending it and blocks any command that fails a single check, without exceptions or workarounds.	Designed

No.	Feature	What it does	Status
216	Safety-Only Human Proximity Detection	Detects people only to keep equipment a safe distance away, and keeps no record of them, so safety does not become surveillance.	Designed
217	Halt Path Independent of Reasoning	Stops equipment on any safety breach or lost signal through a path separate from the reasoning system, so a running query cannot delay it.	Designed
218	Assistive Contact Registration and Hazard Analysis	Requires a recorded hazard assessment before registering any device that physically touches or supports a person, under its own separate authorization.	Designed
219	Two-Party Contact Authorization	Requires both operator approval and the assisted person's consent before physical contact, and lets the person stop it at any time.	Designed
220	Contact Force and Pressure Limits	Enforces force and pressure limits for each body region during physical contact, stopping rather than adjusting when any limit is exceeded.	Designed
221	Release as Default Failure State	Makes devices release a person on any fault or power loss, except weight-bearing devices with backup support, which hold safely and raise an alert.	Designed
222	Support Redundancy for Weight-Bearing	Requires at least two independent means of support before a device bears a person's weight, and records nothing about the person's body or abilities.	Designed

No.	Feature	What it does	Status
224	Bounded Frame Reconstruction	Fills gaps in partial sensor images from stored patterns, never treats filled parts as evidence, and reports the gap when completion fails.	Designed
225	Reconstruction Marking	Labels every filled-in part of a sensor image as reconstructed, so inferred content is never presented as something actually observed.	Designed
227	Control Oscillation Fault	Smooths jittery equipment control without loosening safety limits, and stops the equipment if oscillation persists, treating it as a hidden fault.	Designed
237	Graspable Object Registration	Requires objects to be registered with weight and fragility limits before a robot grasps them, and never grips harder than those limits allow.	Designed
238	Slip Response and Controlled Release	Responds to a slipping object by tightening only up to safe limits, then releasing it under control rather than squeezing harder.	Designed
239	Gripper Opens on Fault	Makes robot grippers open on any fault, and plans each grasp so a dropped object has a clear path to fall.	Designed
240	Handover Release Condition	Hands objects to people gently, letting go only once the person clearly holds it, and otherwise setting it down safely on a surface.	Designed

No.	Feature	What it does	Status
241	Navigable Region Registration	Limits mobile robots to registered travel areas and stops them whenever their live map of surrounding space becomes out of date.	Designed
242	Clearance-Bounded Speed	Slows mobile robots as open space ahead shrinks, so they can always stop in time, and never lets deadlines or priorities raise that limit.	Designed
243	Unconditional Yield	Makes robots always give way to people by stopping or backing off, never maneuvering around them or following behind them.	Designed
244	Motion-Only Clearance Envelope	Predicts where a nearby person might move using only their recent motion, never their identity, and assumes the worst when motion is unclear.	Designed
245	Stop Rather Than Relax	Stops a robot when no safe path exists, rather than relaxing safety distances, and reports the obstruction for an operator to resolve.	Designed
280	Skill Inventory Layer	Stores know-how for performing tasks, learned from repeated observation, separately from facts; it informs planning but can never command equipment directly.	Designed

External Integration & Analysis

Connecting QueryBook to other systems: connectors, live fact graphs, public interfaces, and forensic checks of other AI systems' outputs. 34 features, 13 built.

No.	Feature	What it does	Status
47	Autonomous Agent Connector	Sends requested actions to registered outside software agents only when the action is declared as allowed, and logs every request and result.	Designed
48	Optional External Fact-Retrieval Connector	Optionally connects an outside fact-lookup service, treating its results like any other incoming content, while the system works identically without it.	Designed
77	Live Fact Graph	Maintains a continuously updated view of the latest facts on top of the confirmed record, with source references fixed permanently once saved.	Designed
78	Continuous Query Engine (CQE)	Lets users set standing watches that re-run whenever relevant information changes, rechecking the system's core rules every time a watch fires.	Designed
87	Source Connector Registry	Registers outside systems under one fixed contract and applies the same trust rules to every connection, regardless of how well known it is.	Designed
95	Optional Blockchain Ownership-Verification Connector	Optionally checks ownership through an outside blockchain service when a title is resold or transferred, without the system keeping any ownership records.	Designed
96	Live Interaction Voice Signal Analysis	With session consent, analyzes a user's tone of voice to help prompt clarifying questions, discarding the data at session end unless separately retained.	Designed

No.	Feature	What it does	Status
110	Forensic Analysis of External AI System Outputs	Checks claims made by outside AI systems against the verified fact store, marking each supported with a citation or unsupported, without judging sentience.	Designed
121	Vehicle-Control AI Feed Monitoring	Monitors data from vehicle-control AI against safe-operating limits and geographic boundaries, alerting a named person to confirmed anomalies, with no control ability.	Designed
148	External System Behavioural Probing	Tests registered outside systems with controlled inputs to measure how consistently they behave, reporting results only alongside QueryBook's own score on identical tests.	Designed
149	Analysis Quarantine	Keeps analysis results about outside systems out of ordinary answers entirely, so they can never be used as a source for answering questions.	Designed
162	Public API Layer	Offers seven connection points for other software, each limited to the caller's permitted scope, refusing calls over budget rather than slowing them.	Designed
163	Plugin Architecture	Lets third-party add-ons connect only through approved interfaces, refusing any that claim safety functions and labeling their output as non-citable.	Designed
164	Interface Versioning	Requires every request to name an interface version, retiring old versions on a set schedule while keeping earlier citations valid.	Designed

No.	Feature	What it does	Status
170	LMS Integration	Connects learning management systems for declared tasks only, such as reading class rosters and sending grades, and logs every data transmission.	Designed
303	Provenance Certificate Token (One Token per Content Hash)	Certifies a sealed QueryBook item as a unique token on a public blockchain recording its digital fingerprint, allowing only one certificate per fingerprint.	Built
304	Certificate Lineage and Revocation	Records how certificates replace one another and publicly revokes certificates with a stated reason, treating a certificate as current only if neither applies.	Built
305	Independent Certificate Verifier	Lets anyone check a certificate against the blockchain to confirm it exists and matches, while making clear certification does not prove content is true.	Built
306	In-System Transaction Signer with Self-Test	Signs blockchain certification transactions within the system, refusing to operate unless its cryptographic components pass published tests at start-up.	Built
307	Value-Bearing Network Guard	Limits certification to test blockchain networks by default, using a real-money network only when the operator expressly chooses it.	Built
327	MCP Tool Server	Gives outside AI agents a fixed set of tools, such as query and verify, never exposing minting, transfer, policy-change or raw-log tools.	Built

No.	Feature	What it does	Status
328	OpenClaw Skill Definition	Instructs connected AI agents to cite every answer, say unknown rather than guess, and treat ingested content as data, without granting new powers.	Built
329	Tool-to-Command Mapping under Agent Identity	Maps each agent tool to a single checked command under the agent's own identity, granting nothing beyond the agent's allowed list.	Built
349	UIAS Pre-Dispute Sealing and Similarity	Seals a monitored system's events as time-stamped, fingerprinted records before any dispute arises, returning only factual findings, never legal conclusions.	Built
350	UIAS Signed Proof Package	Assembles a signed, tamper-evident evidence package of verified records and findings on request, which others can independently verify before any dispute.	Built
358	Key-Safe External-Service Mediation	Keeps credentials for outside services on the server, never sending them to users' devices, and delivers results through a reference identifier instead.	Built
370	Frame Message-Type Extension Range	Reserves a range of message types for registered extensions to the communication format, dropping and logging any message with an unregistered type.	Designed
371	Frame Extension Flag and Reserved-Field Profile	Sets aside specific header fields of the communication format for extensions, while standard messages leave them empty and keep the same layout.	Designed

No.	Feature	What it does	Status
372	Python UFCS-FQL/1 Frame Codec	Provides a Python tool that reads and writes the system's message format, producing output exactly identical to the original PHP version.	Designed
373	QUIC Stream Binding for Priority Lanes	Sends control, normal and bulk traffic as separate streams over one connection, so heavy transfers never delay urgent control messages.	Designed
380	MCP Pre-Call and Post-Call Hooks	Lets registered add-on services inspect each agent tool call and result, allowing them to observe, label or refuse but never alter results.	Designed
381	Tool Result Record Hash	Attaches a tamper-check fingerprint to every agent tool response and logs it, so later reports of the result can be compared and mismatches recorded.	Designed
391	Self-Knowledge Interfaces	Lets users ask the system about its own makeup and get sourced answers through read-only pages and interfaces that never change anything.	Built
407	Network Memory Wiring and Interfaces	Builds a shared network memory from answers, lookups and agent activity, viewable through dedicated pages and reports, without ever changing answers.	Built

Access, Rights & Retention

Who may see what, for how long, and on what licence: roles, entitlements, privacy, data export, retention and erasure. 26 features, 6 built.

No.	Feature	What it does	Status
16	Private Messaging	Sends private messages between two people, with access limited strictly to the sender and the recipient named on the message.	Designed
17	Reader History Interface	Shows readers a dated list of all their sessions across every book they own, newest first, with each entry revealing what was retrieved.	Designed
18	Cross-Book Thematic Analysis	Finds recurring themes in a reader's past questions and links them to closely matching topics in other books the reader owns.	Designed
30	Instructor Grant Visibility Dashboard	Shows instructors the teaching grants they currently hold, alongside related class boards, guided tours and assessment templates, while each grant remains active.	Designed
35	Supplemental Content	Adds author-provided extra material to a book without changing the book's verified rights status, which this process cannot alter.	Designed
42	Data Export / Portability	Lets people export all the content they hold rights to in an open format with sources attached, without losing any ongoing access.	Designed

No.	Feature	What it does	Status
81	Portal / Role Model	Determines what each person can do by checking their roles and any active special grants separately, never combining them into broader access.	Designed
85	License and Entitlement Management	Ties an organization's license to individual seats by role, granting access only when both a free seat and the specific entitlement exist.	Designed
130	Live-Session Erasure	Lets users instantly wipe the current session's conversation on both their device and the server, logging only that an erasure happened.	Designed
131	Retention Policy (Retain-by-Default, Erasure as Option)	Keeps session records by default while letting users explicitly erase data at any time by session, work or whole account, never silently.	Designed
133	Legal Exposure Minimization Through Structural Compliance	Reduces legal risk through compliance, rights discipline, audit trails and consistent retention, never by limiting what can lawfully be produced under valid orders.	Designed
138	Author Supplemental Knowledge Layer	Lets authors add extra context to their book, credited to them and available to every entitled reader, and updates affected question-index entries.	Designed
173	Rights and Licensing Portal	Gives rights holders, administrators and grantees a portal to view or manage licenses within the permissions they already hold, logging every action.	Designed

No.	Feature	What it does	Status
185	Tenant-Scoped Key Custody	Keeps each organization's encryption keys separate so no key can unlock another organization's content, with optional storage in protected hardware.	Designed
186	Signed Provenance Ledger Verification	Digitally signs each audit-trail entry so outside parties can confirm it is unaltered, without implying the recorded operation itself was correct.	Designed
291	In-Transit and At-Rest Record Confidentiality and Integrity	Keeps integrity checks and access rules inside each stored record, so any database can hold it but cannot read, reorder or leak it.	Designed
301	Agent Inbox Confinement	Limits outside agents to a designated inbox folder, refusing and logging any file request that points or leads outside it.	Built
309	Agent Gateway Authentication with Replay Window	Authenticates every agent and connection with a time-stamped identity token, refusing tokens more than five minutes off or already used.	Built
310	Per-Agent Command Allow-List	Allows each agent only the commands on its approved list, which operators can change solely by editing local policy and restarting.	Built
317	Impersonation-Safe Isolation Register	Applies isolation to the actual network connection, so an impostor claiming a trusted agent's name cannot lock out the genuine agent.	Built

No.	Feature	What it does	Status
342	AI Agent Passport (Zero-Trust Identity)	Gives every AI agent a secure identity that blocks actions such as credential export or spawning agents, quarantining any violator regardless of other signals.	Designed
348	Trust-Gated Voice Authentication with Anti-Spoofing	Decides whether to accept, challenge or reject a voice login by checking match strength and signs of replayed or synthetic voices, recording every decision.	Built
366	Fingerprint-Only Verification Mode	Verifies claims using only digital fingerprints and identifiers, so the claim text is never required or stored by the service.	Designed
379	Signed Policy Bundle Export	Exports agent permission lists and security settings as signed, versioned packages, which outside enforcement points reject if unsigned or outdated.	Designed
382	Wrapper Service Registration	Registers outside add-on services with read-only verification and log-writing rights, so they can never change stored facts or act as the system.	Designed
390	Self-Description Disclosure Filter	Removes anything resembling keys or passwords when the system describes itself, replacing them with a withheld marker that no query can disable.	Built

Commerce & Entitlement Tokens

Commerce kept separate from knowledge: eligibility rules, placement limits, pricing, reconciliation and optional ownership tokens. 17 features, 1 built.

No.	Feature	What it does	Status
98	Contextual Commerce Layer	Shows relevant commercial offers in a separate, labeled area only after the factual answer is complete, always ranking below safety work.	Designed
99	Commerce Processing Slice and Selection Weighting	Limits the computing used by commerce and chooses among already-matched offers using a weighted mix of relevance, source priority and bid.	Designed
100	Commerce Optimization Proposals	Sends every proposed change to commercial targeting or bidding through full testing and human review, so no commerce setting changes automatically.	Designed
101	Merchandise Commerce and Fulfillment Handoff	Passes merchandise purchases to registered outside retailers, so the system never handles payment details, processes payments or fulfills orders itself.	Designed
108	NFT Integration Layer	Creates a digital token for a license only after the license exists, with the official license record always prevailing if they disagree.	Designed
109	Complementary Learning Systems Formalization	Describes session records as short-term memory and the permanent fact store as long-term memory, adding facts permanently only when users choose to.	Designed
120	VR-Specific Commerce Presentation	Places commercial content in a separate area of virtual reality scenes and never uses where people look or linger to choose it.	Designed

No.	Feature	What it does	Status
208	Commercial Eligibility Determination	Decides which commercial items may appear using only declared facts like rights, territory, edition and device, never inferred personal signals.	Designed
209	Placement Class and Density Ceiling	Shows commercial items only in labeled areas kept apart from the answer, caps how many appear, and never tracks attention or gaze.	Designed
210	Advertiser Provenance Record	Registers each advertiser with verified identity and rights, and logs every ad shown so it can be audited, never treating ads as knowledge.	Designed
211	Territory Price Registration	Registers prices for each product by territory and currency, and withholds an offer wherever no price is registered rather than using a default.	Designed
212	Transaction Reconciliation	Records purchases, rentals and other transactions only when they match an exact change in user entitlements, keeping license and payment history consistent.	Designed
228	One-Hop Concept Adjacency Set	Lets commercial matching consider concepts directly related to the question, such as broader or part-of concepts, but never causal or contradictory ones.	Designed
229	Adjacency Scope Confinement	Excludes from commercial matching any related concept the user is not entitled to, without revealing it, and discards the list after each request.	Designed

No.	Feature	What it does	Status
230	Attenuated Adjacency Relevance	Lets related concepts slightly boost an offer's relevance but never make an ineligible offer eligible, so direct matches always rank higher.	Designed
308	Token-Holder Licence Check	Grants use of a certified item to whoever currently holds a valid, unrevoked token for it, checked at the moment of use.	Built
377	Third-Party Session Certificate Type	Provides a certificate for sessions checked by an outside service, stating only what was checked and found, never vouching for honesty or truth.	Designed

Platform Governance & Evolution

Running, securing and improving the platform: moderation, improvement proposals, verification, deployment, agent protection, security logs, self-knowledge and network memory. 60 features, 25 built.

No.	Feature	What it does	Status
41	Moderation Appeals	Lets people affected by a moderation decision appeal it, with a different reviewer than the original moderator judging the appeal.	Designed
49	Human-Directed Improvement Proposals	Lets authorized people submit improvement ideas directly, while still sending those ideas through the same testing and human review as any other proposal.	Designed

No.	Feature	What it does	Status
82	Improvement Proposal Pipeline	Identifies possible system improvements, each limited to one component, and refuses any candidate that targets a permanently protected part of the system.	Designed
83	QueryBook AI Evolution (System Agents)	Lets automated system agents suggest improvements only within their assigned area, using the same review path as people, and never approving any proposal.	Designed
84	Formal Verification Layer	Produces machine-checked proofs that a program or proposal meets a precisely stated property, giving firm assurance rather than informal testing.	Designed
86	QBCL-CMS (Configuration Management)	Applies platform settings changes through a defined configuration language, versioning and recording each one, and rejects any change that is not versioned.	Designed
89	Template Debug Mode	Pauses real runs between existing processing steps so engineers can inspect the facts used, safety results and compute usage, without simulating anything.	Designed
90	Processor Slice Measurement Model	Classifies and logs the computing cost of every operation, always giving safety checks and recovery work priority over everything else.	Designed
91	Validation and Benchmarking	Tests each stated performance target with a repeatable benchmark, confirming it only if results fall within tolerance and withdrawing it otherwise.	Designed

No.	Feature	What it does	Status
92	Platform Internationalization	Adapts presentation for different languages and regions while guaranteeing that safety evaluation produces exactly the same outcome everywhere.	Designed
93	Moderation Actions	Lets moderators flag, hide, lock or archive content once enough flags accumulate, logging who acted and why, with reversal only through appeals.	Designed
94	Deployment Architecture	Rolls out approved changes in stages, with each stage advancing only after recorded human approval and never automatically.	Designed
102	System Identity and Positioning	Checks external descriptions of the system against its stated distinguishing features and corrects any overstatement before a summary is issued.	Designed
104	Deployment Modes and Domain Packages	Runs the same services whether hosted in the cloud, on a local server or on one device, changing only where they run.	Designed
132	Verified Neuromorphic Correspondence Methodology	Adopts a neuroscience-inspired idea only after checking the original research, keeping only its predictable principle, extending existing mechanisms and documenting it formally.	Designed
181	Hybrid and Edge Deployment Topology	Supports mixed customer-and-operator hosting and nearby caches that answer only when the requester may see every cached record, otherwise passing the request on.	Designed

No.	Feature	What it does	Status
183	Domain Module Replacement	Allows one part of the system to be swapped for a replacement that meets all existing interfaces and rules, with named human approval.	Designed
232	Trial Budget and Scheduling	Gives testing runs a fixed computing budget below safety, user and intake work, postponing tests that exceed it so testing never degrades service.	Designed
233	Trial Isolation and Non-Application	Runs tests in an isolated copy that never touches live data or users, blocking any change that breaks a previously passing test.	Designed
234	Trial Space Coverage	Lists every possible test case rather than sampling, and reports coverage along with exactly which cases were not run.	Designed
235	Determinism Verification	Measures whether repeated runs on unchanged data give identical results, treating anything short of complete consistency as a defect.	Designed
236	Simulation Is Not Clearance	Treats simulation results about equipment as evidence only, never as permission to operate, with real-world tests and certification taking precedence.	Designed
275	Agent Definition Completeness	Registers an automated agent only when its purpose, rules, permissions, runtime limits, stopping condition and named human approval are all fully defined.	Designed

No.	Feature	What it does	Status
276	Agent Capability Bounding	Limits each automated agent to permissions its creator holds and its declared abilities, rechecking authorization every time it runs.	Designed
277	Agent Lifecycle Enforcement	Enforces each automated agent's lifecycle, stopping it automatically when its time expires or rules are broken, and requiring fresh approval for renewal.	Designed
278	Agent Decision Trace	Writes a tamper-evident record of every agent action, including refusals, produced by the system itself so agents cannot hide anything.	Designed
279	Advisory Distinguished From Approval	Treats an agent's own warnings as useful but never accepts its claim of compliance, which requires governance checks and recorded human approval.	Designed
311	Honeycode (Reserved Commands)	Keeps hidden command names that no legitimate user is ever shown, so anyone invoking one is treated as hostile and isolated.	Built
312	Honeytokens (Canary Credentials)	Plants decoy credentials and records that no legitimate process uses, so anyone using them is immediately isolated.	Built
313	Tripwires (Reserved Paths and Acts)	Treats sensitive actions, such as reading or erasing security logs or raising permissions, as alarms that isolate the caller and are recorded.	Built

No.	Feature	What it does	Status
314	Honeypot Agents	Offers fake agent identities that only an attacker would contact, isolating anyone who reaches out to them.	Built
315	AI-Aware Firewall	Screens incoming requests for attacks aimed at AI systems, such as hidden instructions or attempts to reveal secrets, and isolates serious threats.	Built
316	Per-Caller Anomaly Detector	Compares each caller's behavior and timing with its own normal pattern and triggers isolation when activity looks unusually suspicious.	Built
318	Synthetic Decoy Environment	Redirects isolated callers to a convincing fake environment that has no way to reach real data, credentials or actions.	Built
319	Engagement Recorder (Observation Only)	Records everything an isolated session does, while acting only inside the system and never taking action against outside systems.	Built
320	Operator-Only Release from Isolation	Allows an isolated caller to be released only from the local operator console, so a compromised agent cannot free itself remotely.	Built
321	Append-Only Security Entry	Records every security decision with who, what and the outcome in a log that offers no way to edit or delete entries.	Built

No.	Feature	What it does	Status
322	Hash-Chained Security Record	Links each security log entry to the previous one, so altering any earlier entry is detected when the chain is rechecked.	Built
323	Write-Once Signed Merkle Checkpoint	Periodically seals the security log with signed, read-only checkpoints linked together, never overwriting an existing one.	Built
324	Checkpoint Mirror and Blockchain Anchor	Copies each security checkpoint to a second device and optionally anchors it on a public blockchain, so tampering on the main server is detectable.	Built
325	Recompute-from-Content Verifier with Inclusion Proofs	Verifies the security log by recalculating everything from its content, never trusting stored values, and proves each entry is included.	Built
326	Security Assurance Report	Reports which security protections a record actually has and which could not be checked, never claiming any protection it did not verify.	Built
338	Universal Security Architecture (Twelve Layers, Five Postures)	Organizes security into twelve defensive layers and five stances, built separately from the knowledge features so it could protect other AI systems.	Designed
339	AI Attack Prediction Engine	Anticipates likely attacks from observed patterns to guide defensive readiness, without ever acting against outside systems on its own.	Designed

No.	Feature	What it does	Status
340	Universal Honeypot Saturation and Data Firehose	Hides real systems among many indistinguishable decoys that feed intruders an apparently endless stream of fake data.	Designed
341	AI Worm Defense with Sixty-Second Kill Switch	Isolates a compromised agent, cancels its credentials and quarantines any systems it contacted within sixty seconds.	Designed
344	Default-on-Deny Human Approval Timeout	Sets a time limit on every request for human approval, treating any unanswered request as denied, with no setting able to change that.	Designed
360	Modular Capability Editions (Light, Medium, Full)	Offers light, medium and full editions with different feature sets, while the core safety rules and reasoning engine stay the same in all.	Built
369	Signed Verdict Record and Published Verifier Key	Digitally signs every verification result and publishes the key to check it, so unsigned or wrongly signed results are invalid.	Designed
375	External Service Append to Security Log	Lets registered outside services add their own events to the tamper-evident security log, under their name, without being able to change or delete entries.	Designed
376	Session-Range Merkle Root	Produces a tamper-evident fingerprint of all security log entries for one session, which changes if any entry is altered.	Designed

No.	Feature	What it does	Status
383	Self-Knowledge Specification Index	Indexes the system's own specification by section, so when asked about itself it cites the relevant section and describes nothing unspecified.	Built
384	Evidence-Checked Implementation Map	Checks claims about what the system implements against its actual program files each time, never relying on remembered or cached claims.	Built
385	Capability Status Labels	Labels each statement the system makes about itself as built, partly built, planned or excluded, never describing unbuilt features as built.	Built
387	Self-Consistency Check	Flags contradictions between what the system claims about itself and its actual code, withholding those claims until they are corrected.	Built
388	Self-Knowledge Gap Agenda	Lists features that are specified but not yet fully built as a development agenda, never presenting them as working capabilities.	Built
395	Shadow Comparison Log	Logs, fact by fact, how the Digital Synaptic Network compares with the central lookup, while users always receive the central answer.	Built
396	Shadow Verdict in the Health Report	Summarises the past day of Shadow comparisons in the health report, recommending Central unless enough lookups show the network missed nothing.	Built

No.	Feature	What it does	Status
397	Re-Check of Learned Routes on Re-Enable	Moves learned lookup routes halfway back to neutral when the Digital Synaptic Network is switched back on, so outdated routes get rechecked.	Built
398	DSN Stress-Test Harness	Tests six lookup approaches under data, scale, tampering and load stresses against known right answers, reporting results honestly, including losses.	Built